

*Continuing the debate: a response to a
response – persistent problems and
practices in Information Systems
Development as enduring contradictions of
new software development approaches?*

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In 'The enduring contradictions of new software development approaches: A response to "Persistent Problems and Practices in Information Systems Development"', Baskerville *et al.* (2007) discuss our work as presented in Kautz *et al.* (2007). They interpret our research as offering 'controversy over the degree to which the set of software development practices, variously called agile methodology, Internet speed methodology, and short cycle time development, are different from more traditional or plan-based development approaches' and as leading the reader 'to wonder how history might regard software development agility'. This is a noteworthy interpretation of our work. Like Baskerville *et al.*, we welcome the resurrected tradition of a written academic debate, and will use the opportunity thankworthy given to us by the editors of ISJ to clarify the contents and intent of our original publication.

The aim of our paper is to analyse the problems facing Information Systems Development (ISD), and the practices applied by developers and managers to cope with these problems as they emerge in different contexts and at different points in time, i.e. 'traditional' ISD as performed in the early 1990s and 'web' ISD as performed in the beginning of the century. Thus, we use contemporary literature on Web development as an example of research that argues a need for new concepts, methodologies and tools as a result of fundamental changes in ISD. Our point is that such changes may be obvious on the surface level, but the problems and conditions facing contemporary practice are not, in their essence (Brooks, 1987), different from the challenges previously experienced. Likewise, the practices – which are not the same as methods or methodologies – that are used to solve the problems are not essentially different from previously observed ISD practices. This can be illustrated with an example: programming in .Net or Visual Basic is of course different from programming in Assembler. It is, however, still programming, i.e. solving information processing problems by means of executable code.

In their response, Baskerville *et al.* put much emphasis on the concept of agility, and contrast the characteristics of Web development with agile development. The comparison is most interesting, but we reiterate that our analysis and interest concerns ISD practices – past and

present – and that the Web development literature is used as an example and a starting point. The discussion of agile development, introduced by Baskerville *et al.*, is, however, very relevant, and opens the door for new research into how and why agility became a fashionable ISD practice and a prominent subject of ISD research. Agile development shares several of the characteristics of past and present ISD practice that we describe in the paper, and may therefore prove to be a more appropriate foundation for conceptualizing and guiding ISD than ‘traditional’ ISD methods.

What appears to be new, however, is the changed view on ISD practice and how ISD should be performed in practice as demonstrated by the excitement over agile development: from being the target of ridicule and scorn by proponents of ‘traditional’ ISD methods, common ISD practices are now labelled ‘agile development’ and are becoming increasingly canonical. Still, opposite and influential forces are pulling in the direction of more formalization of ISD through the use of methods and standardizations of processes (Fitzgerald *et al.*, 2002). Such different positions on how ISD should be carried out illustrate the need for a theoretical foundation from which to understand and support practice.

In addition to empirical research about new ISD phenomena and the changing perceptions here of as they appear at the surface level, we therefore advocate a need for conceptual analytical research (Järvinen, 2004) that synthesize the many individual studies and their often conflicting views and findings into a deep understanding of the underlying problems and coping practices of ISD. We argue that a focus on the contextual and temporal similarities of ISD practices and problems – rather than the differences – that can be found in the already rich and comprehensive body of knowledge on ISD allows for the development of a firm theoretical foundation from which to understand and assess ISD problems, practices, and formalized methods. We hope that our work on persistent problems and practices will stimulate further discussion on and development of a deep, more theoretical understanding of ISD, and we look forward to continuing the debate in and outside ISJ.

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